Observability Stack & Monitoring Infrastructure

📖 Overview

This document details the deployment and configuration of the **observability and monitoring infrastructure** using **Grafana** and **Prometheus** on a dedicated Ubuntu Server 24.04 system. The monitoring stack is strategically deployed within the **Monitoring VLAN (VLAN 60)** to provide centralized visibility into the entire lab infrastructure while maintaining network segmentation and security isolation.

🎯 Deployment Objectives

Primary Goals

- Establish comprehensive infrastructure monitoring capabilities
- Deploy Grafana for visualization and dashboarding
- Implement Prometheus for metrics collection and storage
- Create foundation for observability across all lab VLANs
- Provide centralized monitoring without network interference

Strategic Architecture

- **VLAN Assignment**: Monitoring VLAN (VLAN 60 - `192.168.60.0/24`)
- **Network Isolation**: Dedicated monitoring segment separated from operational systems
- **Centralized Approach**: Single monitoring stack serving entire lab infrastructure
- **Scalable Design**: Foundation for future monitoring expansion

🖥️ Server Deployment & Network Configuration

Hardware & System Specifications

Component	Details
Hostname	`nbl-core-ub01`
Operating System	Ubuntu 24.04 LTS Server
Architecture	x86_64
Network Interface	Primary ethernet interface

Component	Details
Switch Connection	Port 8 (Untagged VLAN 60)
Physical Location	Dedicated standalone system

Network Configuration Strategy

Switch Port Assignment

- **Physical Port**: Port 8 on TP-Link TL-SG108E managed switch
- **VLAN Configuration**: Untagged member of VLAN 60 (Monitoring)
- **Network Segment**: `192.168.60.0/24` subnet
- **Gateway Assignment**: `192.168.60.1` (pfSense)

Static IP Implementation

For consistent monitoring service accessibility, a static IP configuration was implemented:

Target Network Configuration: | Parameter | Value | |-----------|-------| | **IP Address** | `192.168.60.2/24` | | **Gateway** | `192.168.60.1` | | **DNS Servers** | `8.8.8.8`, `8.8.4.4`, `1.1.1.1` | | **Network Interface** | Primary ethernet |

Network Configuration Process

Ubuntu 24.04 Server uses Netplan for network configuration management:

```
# /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  ethernets:
    [interface-name]:
      addresses:
        - 192.168.60.2/24
      routes:
        - to: 0.0.0.0/0
          via: 192.168.60.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
          - 1.1.1.1
```

Network Configuration Evidence

Connectivity Verification

```
# Test gateway connectivity
ping -c 4 192.168.60.1

# Verify external connectivity
ping -c 4 8.8.8.8

# Confirm DNS resolution
nslookup google.com
```

Verification Results: ✅ **Gateway Connectivity**: Successfully reached `192.168.60.1`
✅ **Internet Access**: External connectivity confirmed via outbound NAT
✅ **DNS Resolution**: Name resolution functional
✅ **VLAN Assignment**: Confirmed placement in Monitoring VLAN

🔐 Inter-VLAN Access Configuration

pfSense Firewall Rule Implementation

To enable administrative access from the Management VLAN and proper monitoring functionality, specific firewall rules were configured on pfSense.

Management to Monitoring VLAN Access

Rule Purpose: Allow administrative access to monitoring systems from Management VLAN

Firewall Rule Configuration: - **Source**: Management VLAN subnet (`192.168.10.0/24`) - **Destination**: Monitoring VLAN subnet (`192.168.60.0/24`) - **Action**: Allow - **Protocol**: Any - **Purpose**: Administrative access and monitoring management

Monitoring to Management VLAN Access

Rule Purpose: Enable monitoring systems to collect metrics from Management VLAN resources

Firewall Rule Configuration: - **Source**: Monitoring VLAN subnet (`192.168.60.0/24`) - **Destination**: Management VLAN subnet (`192.168.10.0/24`) - **Action**: Allow - **Protocol**: Any - **Purpose**: Metrics collection and system monitoring

Outbound Internet Access

Both VLANs maintain their ability to reach external resources for updates, package downloads, and external monitoring targets.

Access Validation

```
# From Management VLAN (192.168.10.2) - test SSH access to monitoring server
ssh nantwi@192.168.60.2

# From Monitoring VLAN - test connectivity to Management systems
ping 192.168.10.1  # pfSense gateway
ping 192.168.10.2  # Ansible controller
```

Results: ✅ Bidirectional connectivity established and verified

📊 Grafana Deployment & Configuration

Installation Process

Grafana was installed using the official repository method for Ubuntu systems:

```
# Update system packages
sudo apt update && sudo apt upgrade -y

# Install required dependencies
sudo apt install -y software-properties-common wget

# Add Grafana GPG key
wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -

# Add Grafana repository
echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee /etc/apt/sources.list.d/gra

# Update package list and install Grafana
```

```
sudo apt update
sudo apt install -y grafana
```

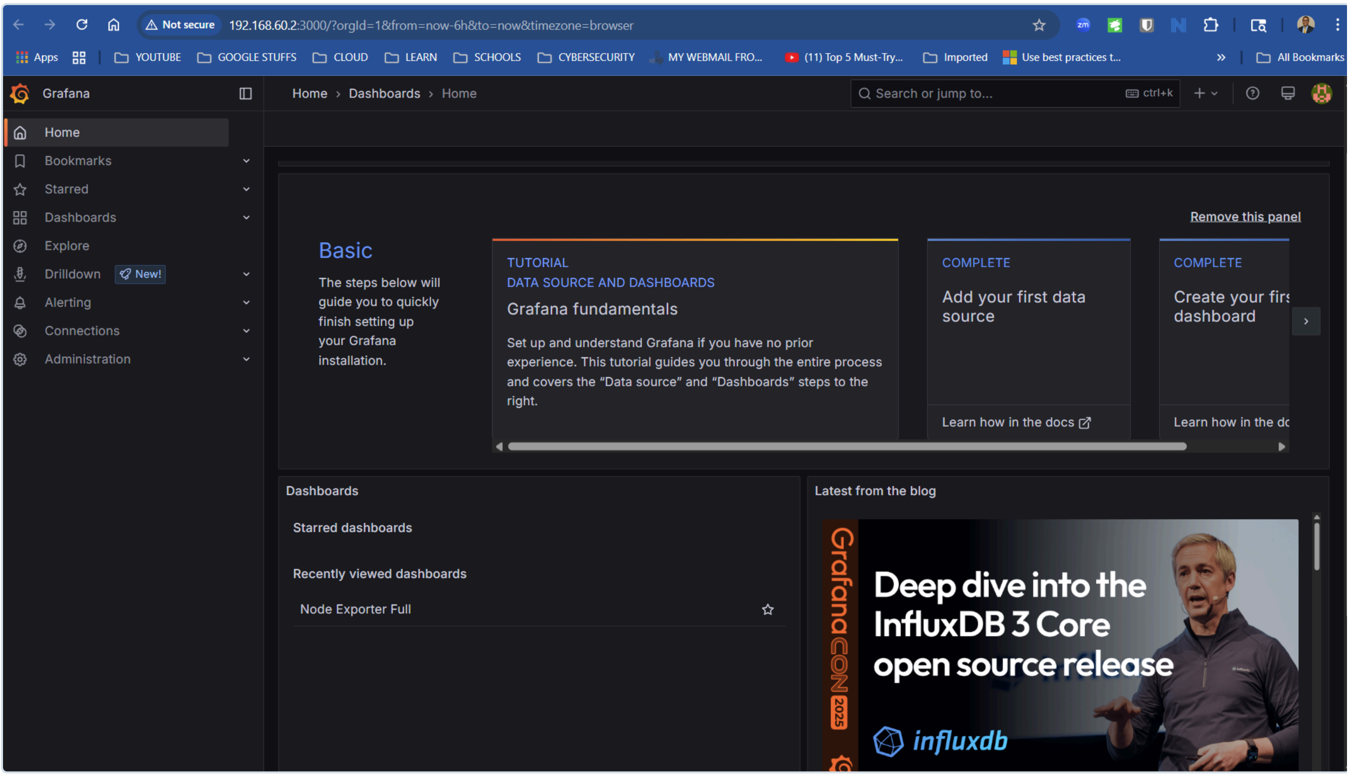
Service Configuration

```
# Enable Grafana service for automatic startup
sudo systemctl enable grafana-server

# Start Grafana service
sudo systemctl start grafana-server

# Verify service status
sudo systemctl status grafana-server
```

Grafana Access Configuration

Service Details: - **Port**: 3000 (default Grafana port) - **Protocol**: HTTP (HTTPS can be configured later) - **Access URL**: `http://192.168.60.2:3000` - **Default Credentials**: admin/admin (changed on first login)

Grafana Dashboard Access



Initial Configuration Validation

✅ **Service Status**: Grafana-server active and running
✅ **Network Accessibility**: Dashboard accessible from Management VLAN
✅ **Web Interface**: Grafana UI responsive and functional
✅ **Authentication**: Login process working correctly

🔍 Prometheus Deployment & Configuration

Installation Process

Prometheus was installed using the Ubuntu package manager:

```
# Install Prometheus from Ubuntu repositories
sudo apt install -y prometheus

# Verify installation
prometheus --version
```

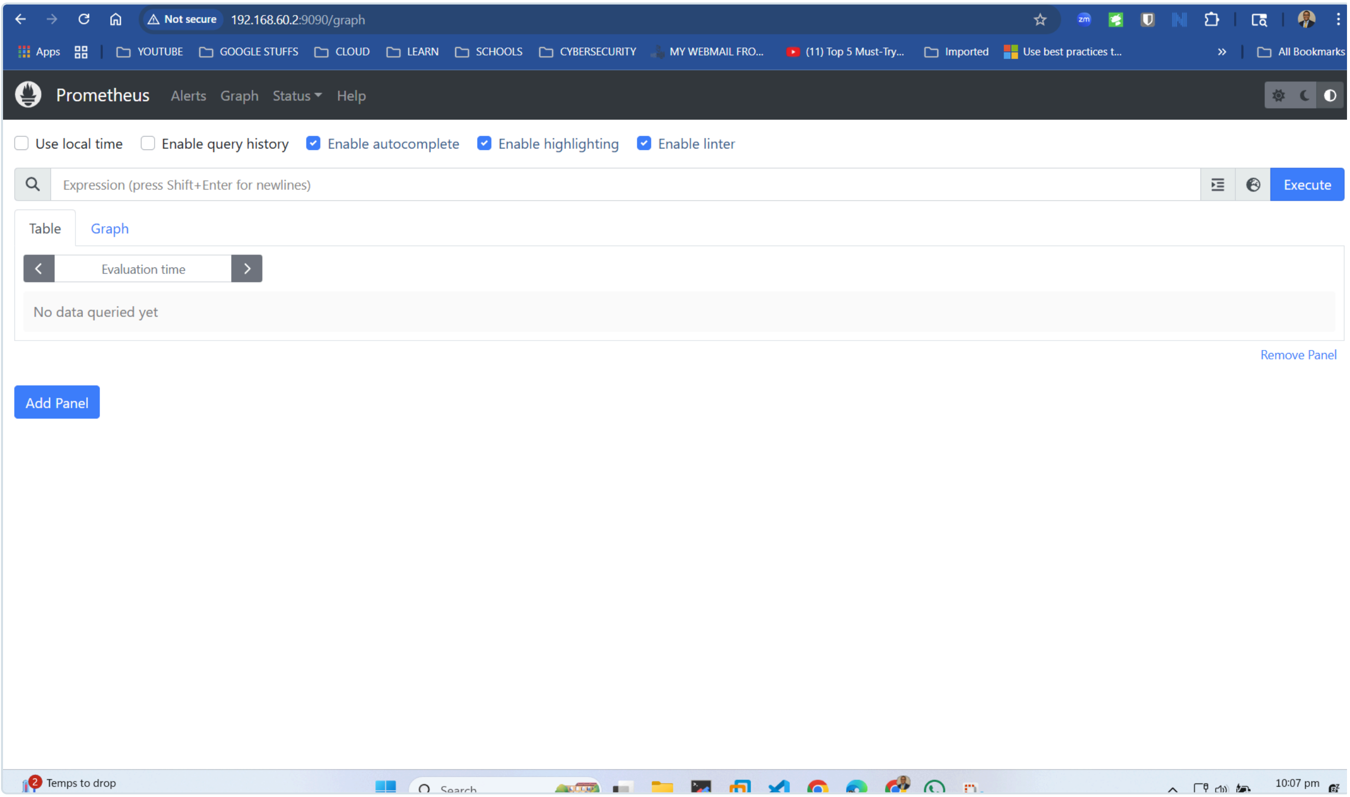
Service Configuration

```
# Enable Prometheus service
sudo systemctl enable prometheus

# Start Prometheus service
sudo systemctl start prometheus

# Verify service status
sudo systemctl status prometheus
```

Prometheus Access Configuration

Service Details: - **Port**: 9090 (default Prometheus port) - **Protocol**: HTTP - **Access URL**: `http://192.168.60.2:9090` - **Configuration**: Default configuration with local metrics

Prometheus Interface Access



Default Configuration Status

- **Data Collection**: Self-monitoring metrics operational
- **Storage**: Local time-series database functional
- **Query Interface**: PromQL query interface accessible
- **API Endpoints**: REST API available for integrations

🛠️ System Integration & Accessibility

SSH Administrative Access

SSH access was configured for remote administration and maintenance:

```
# Ensure SSH service is installed and running
sudo systemctl status ssh

# Test SSH connectivity from Management VLAN
ssh nantwi@192.168.60.2
```

SSH Access Verification: ✅ **Service Status**: SSH daemon active and listening
✅ **Network Access**: SSH accessible from Management VLAN
✅ **Authentication**: Key-based and password authentication functional
✅ **Security**: SSH access controlled by pfSense firewall rules

Service Port Summary

Service	Port	Protocol	Access URL	Status
Grafana	3000	HTTP	`http://192.168.60.2:3000`	🟢 Active
Prometheus	9090	HTTP	`http://192.168.60.2:9090`	🟢 Active
SSH	22	TCP	`ssh nantwi@192.168.60.2`	🟢 Active

🔧 Current Monitoring Capabilities

Grafana Features Available

- **Dashboard Creation**: Visual dashboard development environment
- **Data Source Integration**: Ready for Prometheus and other data sources
- **User Management**: Authentication and authorization system
- **Alerting Framework**: Alert rule configuration capabilities
- **Plugin System**: Extensible with community and commercial plugins

Prometheus Capabilities

- **Metrics Collection**: Time-series data collection and storage
- **Query Language**: PromQL for metric analysis and aggregation
- **Data Retention**: Configurable retention policies
- **API Access**: RESTful API for external integrations
- **Service Discovery**: Automatic target discovery mechanisms

Current Monitoring Scope

- **Self-Monitoring**: Both services monitoring themselves
- **System Metrics**: Basic host-level metrics collection
- **Network Connectivity**: Network reachability monitoring
- **Service Health**: Application and service status monitoring

🚀 Integration Framework

Prepared for Expansion

The monitoring infrastructure is architected to support:

Data Source Integration

- **Additional Prometheus Instances**: Multiple Prometheus servers for different VLANs
- **Log Aggregation**: Integration with log management systems (Loki, ELK)
- **Database Monitoring**: MySQL, PostgreSQL, MongoDB metrics

- **Application Metrics**: Custom application monitoring

Exporter Integration

- **Node Exporter**: System-level metrics (CPU, memory, disk, network)
- **Blackbox Exporter**: Network endpoint monitoring and availability
- **SNMP Exporter**: Network device monitoring (switches, routers)
- **Custom Exporters**: Application-specific metrics collection

External Monitoring

- **Wazuh Integration**: Security metrics and log analysis
- **pfSense Monitoring**: Firewall and network performance metrics
- **Infrastructure Monitoring**: VM and container metrics
- **Service Monitoring**: Application availability and performance

🔐 Security Implementation

Network Security

- **VLAN Isolation**: Monitoring systems contained within dedicated VLAN
- **Firewall Controls**: pfSense manages all inter-VLAN communication
- **Access Restrictions**: Administrative access limited to Management VLAN
- **Service Exposure**: Monitoring interfaces not exposed to internet

Authentication & Access Control

- **Grafana Authentication**: Default admin credentials changed on deployment
- **SSH Access**: Public key authentication configured
- **Service Security**: Services running with appropriate user permissions
- **Network Filtering**: pfSense firewall rules control access patterns

📈 Performance & Scalability

Current Resource Utilization

- **CPU Usage**: Minimal load during idle state
- **Memory Consumption**: Efficient resource utilization
- **Storage Requirements**: Adequate space for metrics retention
- **Network Bandwidth**: Low bandwidth requirements for current monitoring scope

Scalability Considerations

- **Data Retention**: Configurable retention policies for storage management
- **Query Performance**: Prometheus optimized for time-series queries
- **Dashboard Performance**: Grafana efficient for visualization rendering

- **High Availability**: Framework supports future HA implementation

🔄 Operational Status

Service Health Monitoring

```
# Check all monitoring services
sudo systemctl status grafana-server prometheus ssh

# Verify network connectivity
ss -tlnp | grep -E ':(3000|9090|22)'

# Test web interfaces
curl -I http://192.168.60.2:3000  # Grafana health
curl -I http://192.168.60.2:9090  # Prometheus health
```

Connectivity Matrix

Source	Destination	Protocol	Status
Management VLAN	Monitoring Server	HTTP/HTTPS	✅ Accessible
Management VLAN	Monitoring Server	SSH	✅ Accessible
Monitoring Server	Internet	HTTP/HTTPS	✅ Accessible
Monitoring Server	Management VLAN	Various	✅ Accessible

🎯 Future Enhancement Pipeline

Planned Integrations

- **Node Exporter Deployment**: System metrics across all VLANs
- **Prometheus Configuration**: Scraping targets for all lab infrastructure
- **Grafana Dashboard Development**: Custom dashboards for lab monitoring
- **Alerting Implementation**: Proactive monitoring alerts and notifications

Advanced Monitoring Features

- **Log Aggregation**: Loki deployment for centralized logging
- **Distributed Tracing**: Application performance monitoring
- **Custom Metrics**: Application and service-specific monitoring
- **Compliance Monitoring**: Security and compliance metric tracking

✅ Deployment Validation

Functional Testing Results

- ✅ **Grafana Accessibility**: Web interface responsive from Management VLAN
- ✅ **Prometheus Functionality**: Query interface operational and responsive
- ✅ **SSH Connectivity**: Remote administration access functional
- ✅ **Inter-VLAN Communication**: Bidirectional network access verified
- ✅ **Service Persistence**: All services survive system restart

Security Validation

- ✅ **VLAN Isolation**: Confirmed placement in Monitoring VLAN
- ✅ **Firewall Rules**: Appropriate access controls implemented
- ✅ **Service Security**: Services running with proper permissions
- ✅ **Access Control**: Administrative access properly secured

Integration Verification

- ✅ **Network Integration**: Proper integration with pfSense infrastructure
- ✅ **DNS Resolution**: Name resolution functional across network
- ✅ **Gateway Access**: Routing through pfSense operational
- ✅ **Internet Connectivity**: External access for updates and integrations

📋 Current Deployment Summary

Successfully Implemented Components

Component	Version	Status	Access Method
Grafana	Latest OSS	🟢 Operational	`http://192.168.60.2:3000`
Prometheus	Latest Stable	🟢 Operational	`http://192.168.60.2:9090`
SSH Service	OpenSSH	🟢 Operational	`ssh nantwi@192.168.60.2`
Network Config	Static IP	🟢 Operational	`192.168.60.2/24`

Infrastructure Integration Status

- **VLAN Assignment**: ✅ Successfully placed in Monitoring VLAN (60)
- **Network Connectivity**: ✅ Full connectivity with appropriate security controls
- **Administrative Access**: ✅ Management VLAN access configured and tested
- **Service Discovery**: ✅ Ready for monitoring target integration
- **Scalability Foundation**: ✅ Architecture supports future expansion

Observability Stack Status: ✅ Complete and Operational
Next Phase: [Automation Platform Deployment]
